

Unified IT Operations & Multi-Kernel Integration

Hardware Audit, Hypervisor Configuration, & WSL 2 Setup

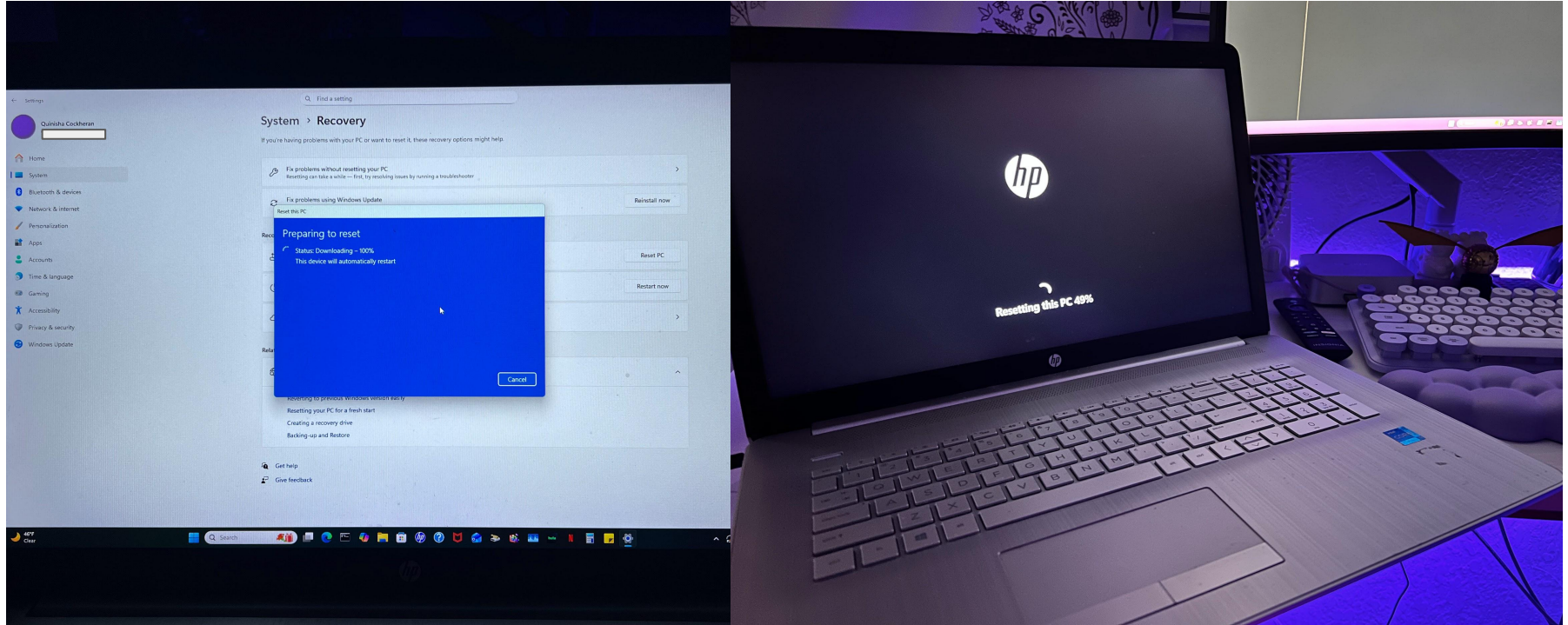
By Quinisha Cockheran

Phase 1: Host Recovery & Hardware Optimization

The following slides consist of photos of Phase 1:

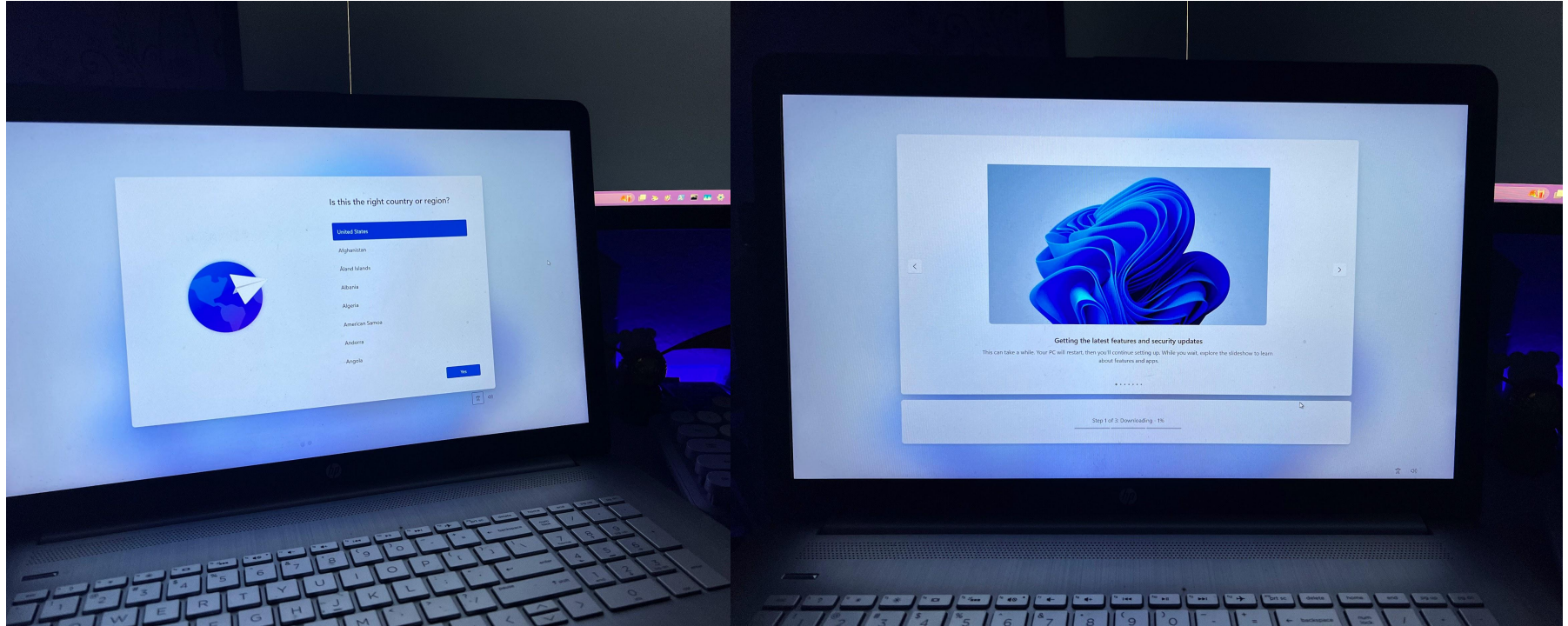
- System Sanitation

System Sanitation



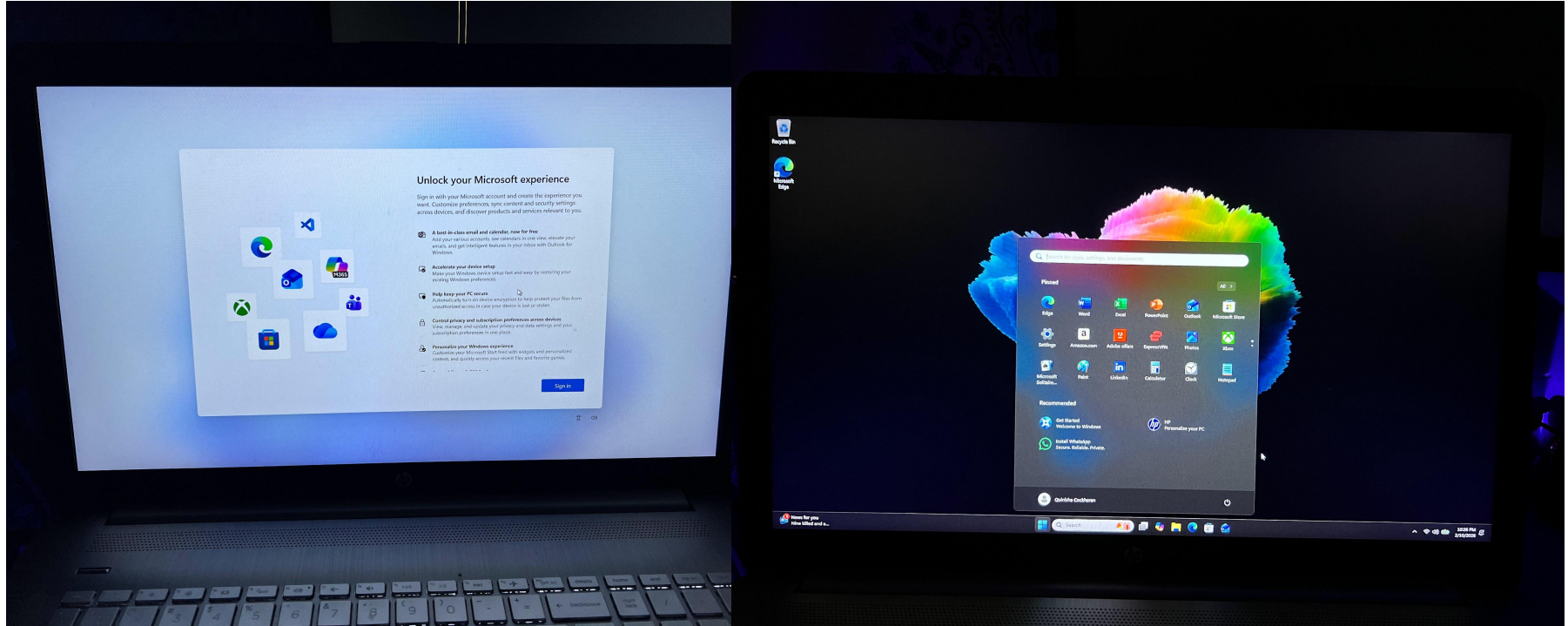
Executed a clean-slate Windows 11 Home deployment to eliminate legacy driver conflicts and kernel-level errors

System Sanitation



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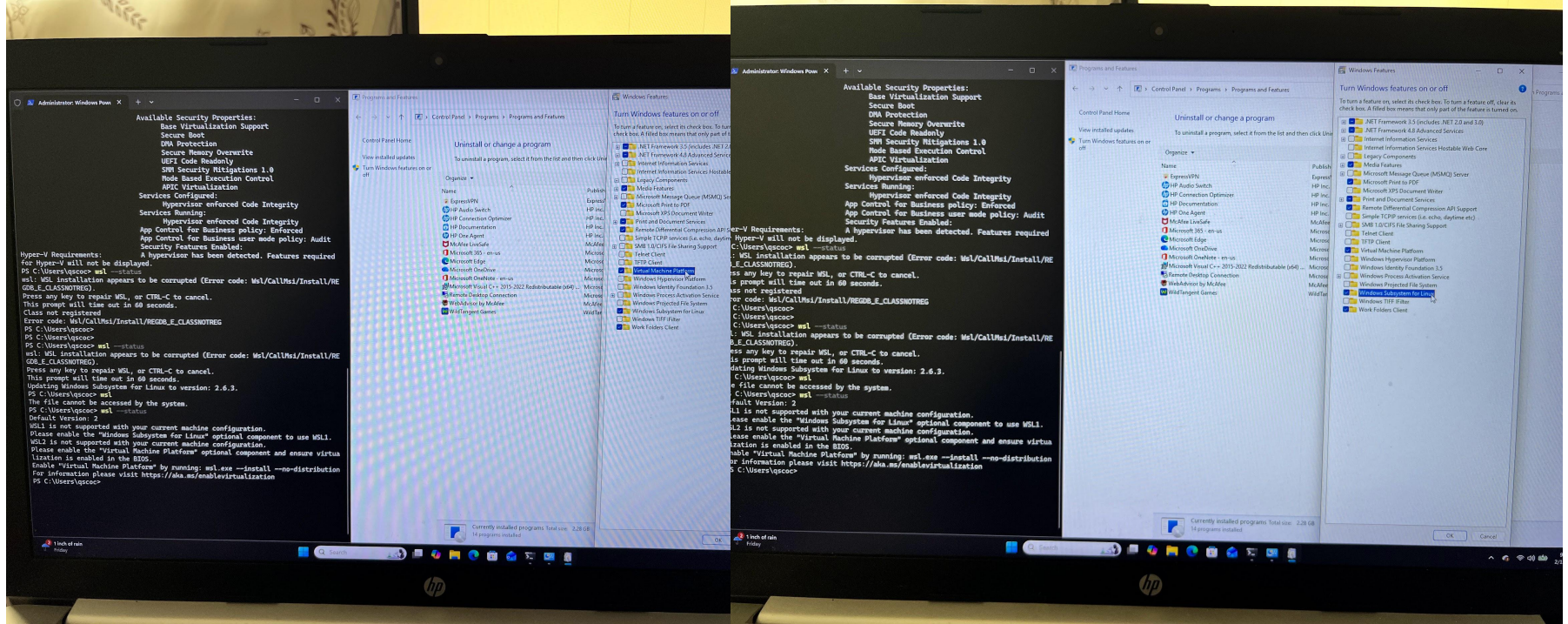
Executed a clean-slate Windows 11 Home deployment to eliminate legacy driver conflicts and kernel-level errors

Phase 2: Native Kernel Integration (WSL 2)

The following slides consist of photos of Phase 2:

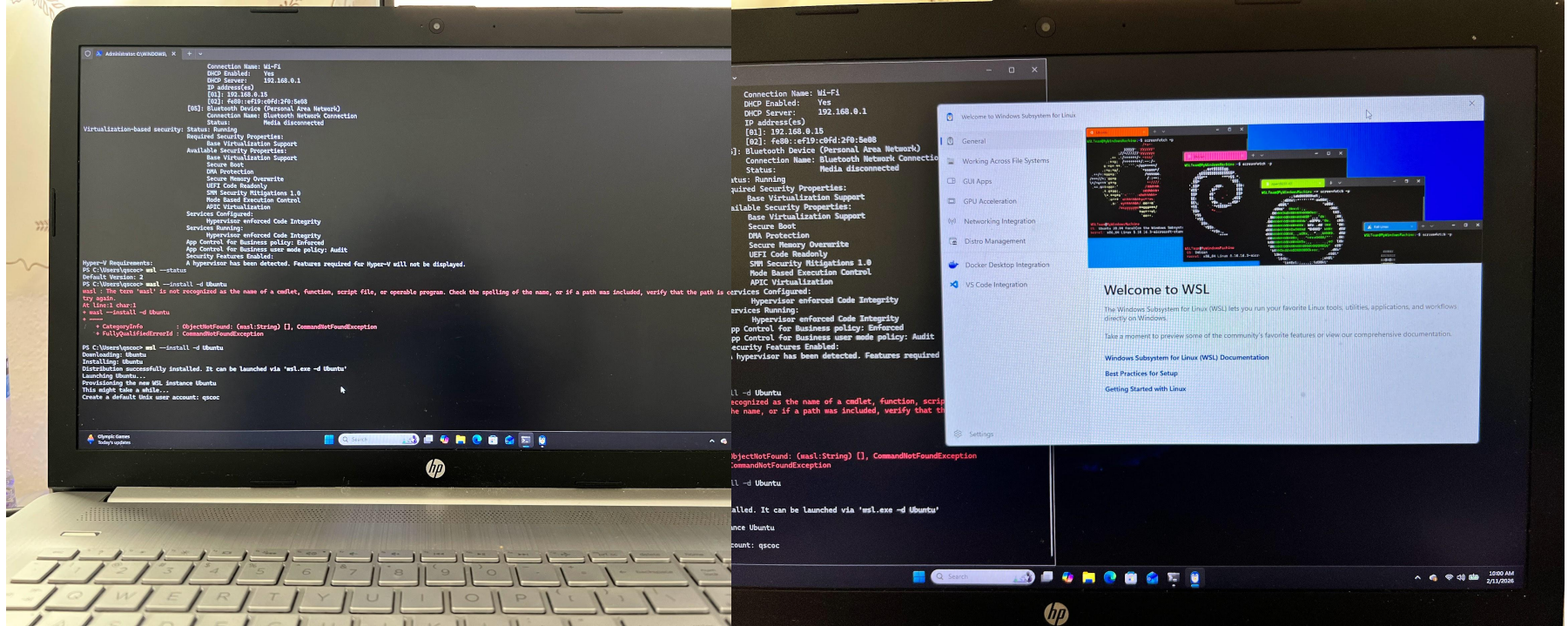
- Backend Initialization
- Environment Deployment
- Verification

Backend Initialization



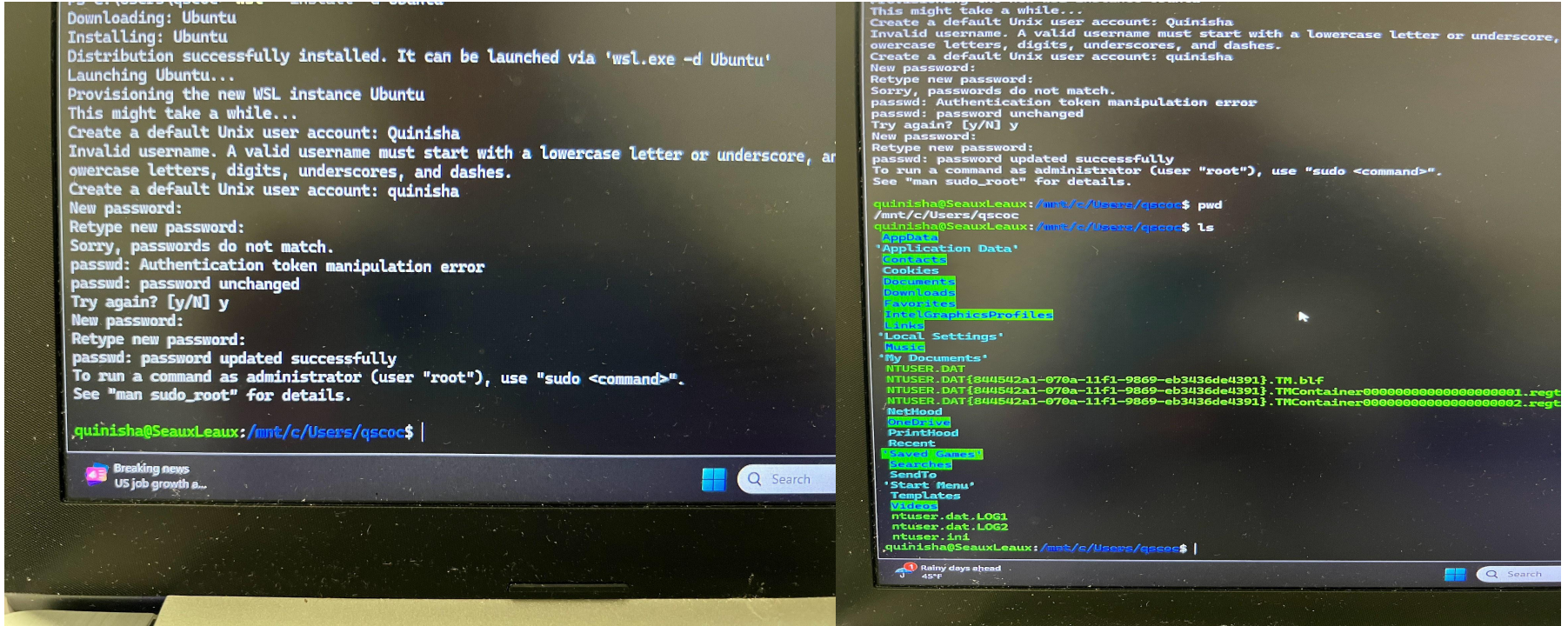
Utilized the Windows Terminal to enable the Virtual Machine Platform and Windows Subsystem for Linux features

Environment Deployment



Successfully installed and deployed Ubuntu 24.04 LTS via the native WSL 2 architecture

Verification



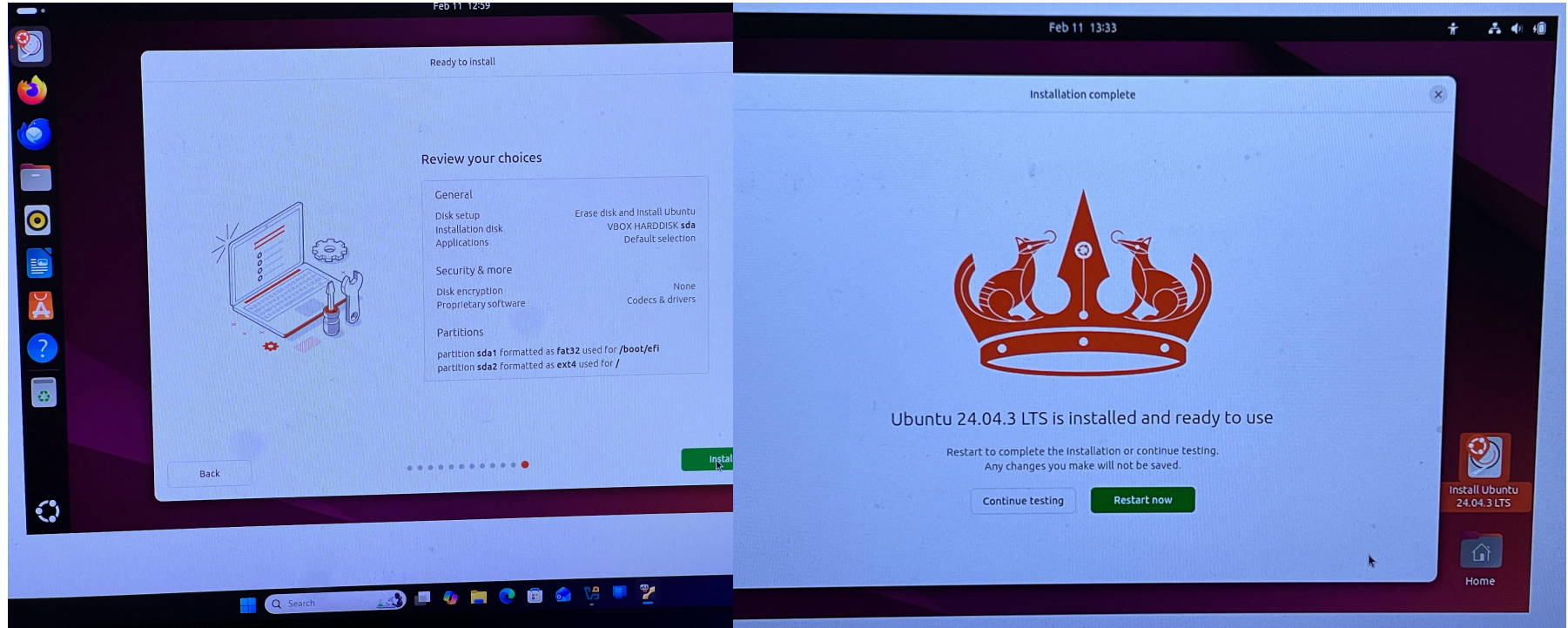
Configured UNIX user credentials and verified shell accessibility, establishing a stable CLI-based Linux environment directly integrated with the Windows host

Phase 3: Full Virtualization & Security Hardening (VirtualBox)

The following slides consist of photos of Phase 3:

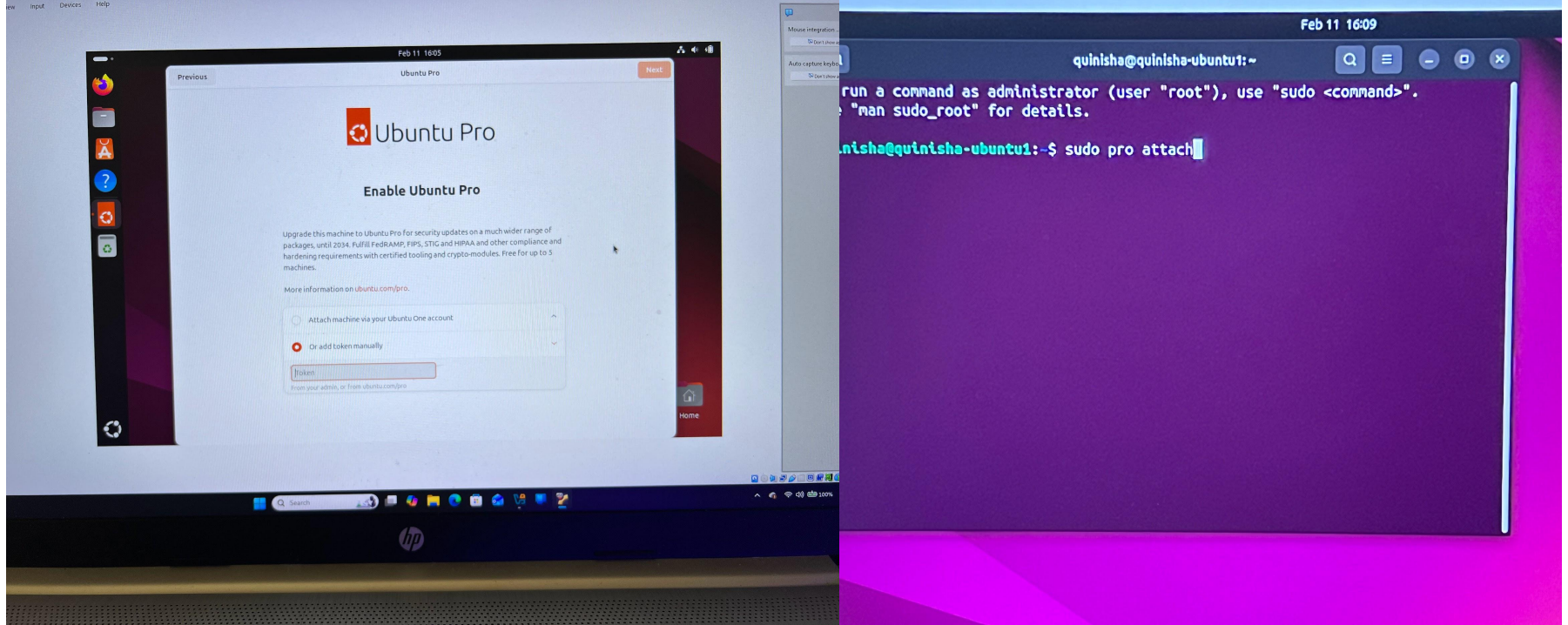
- Hypervisor Setup
- CLI Troubleshooting
- Enterprise Integration

Hypervisor Setup



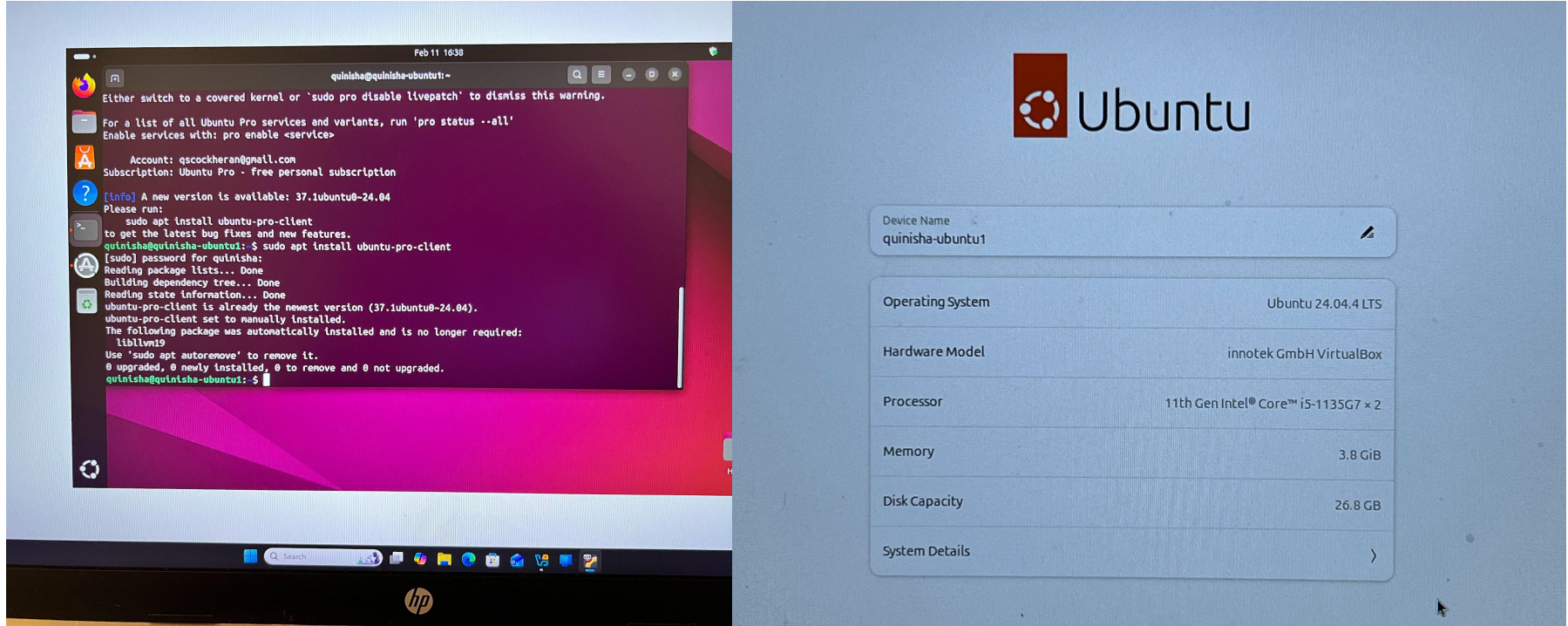
Configured a dedicated VirtualBox instance with 25GB of allocated storage for Ubuntu 24.04.3 LTS

CLI Troubleshooting



Resolved a graphical installer hang during the Ubuntu Pro setup by pivoting to the terminal to manually finalize registration

Enterprise Integration



The image shows two side-by-side screenshots from a Linux desktop environment. The left screenshot is a terminal window titled 'quinisha@quinisha-ubuntu:~' showing the process of installing Ubuntu Pro. The right screenshot is the 'About' window in Ubuntu, displaying system details.

Terminal Output:

```
quinisha@quinisha-ubuntu:~  
Either switch to a covered kernel or 'sudo pro disable livepatch' to dismiss this warning.  
For a list of all Ubuntu Pro services and variants, run 'pro status --all'  
Enable services with: pro enable <service>  
  
Account: qscckheran@gmail.com  
Subscription: Ubuntu Pro - free personal subscription  
  
[info] A new version is available: 37.1ubuntu0-24.04  
Please run:  
sudo apt install ubuntu-pro-client  
to get the latest bug fixes and new features.  
quinisha@quinisha-ubuntu:~$ sudo apt install ubuntu-pro-client  
[sudo] password for quinisha:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
ubuntu-pro-client is already the newest version (37.1ubuntu0-24.04).  
ubuntu-pro-client set to manually installed.  
The following package was automatically installed and is no longer required:  
libllvm19  
Use 'sudo apt autoremove' to remove it.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
quinisha@quinisha-ubuntu:~$
```

Ubuntu System Information:

Device Name	quinisha-ubuntu1
Operating System	Ubuntu 24.04.4 LTS
Hardware Model	innotek GmbH VirtualBox
Processor	11th Gen Intel® Core™ i5-1135G7 × 2
Memory	3.8 GiB
Disk Capacity	26.8 GB
System Details	>

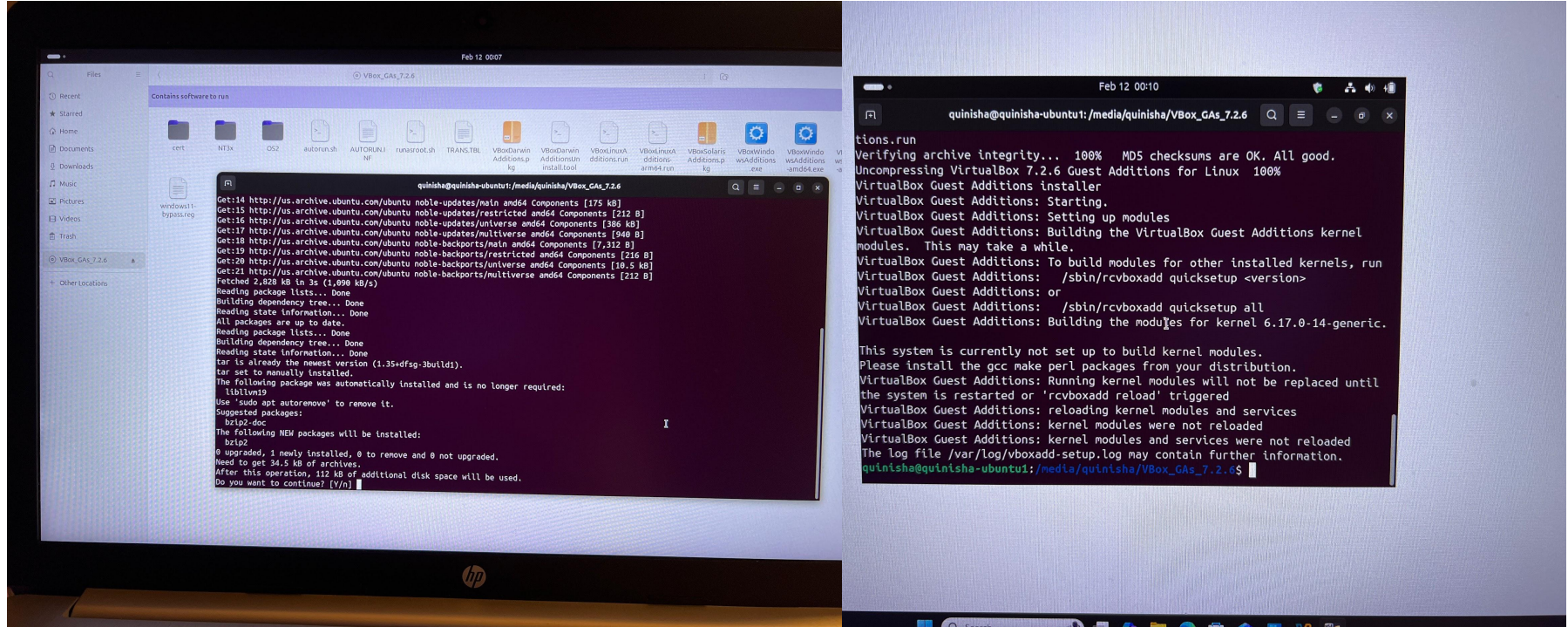
Manually attached the system to Ubuntu Pro using a personal token via terminal - Enabled ESM and Livepatch for continuous security patching - Successfully executed a 105.4 MB system update cycle through the Software Updater

Phase 4: Workspace & Display Optimization

The following slides consist of photos of Phase 4:

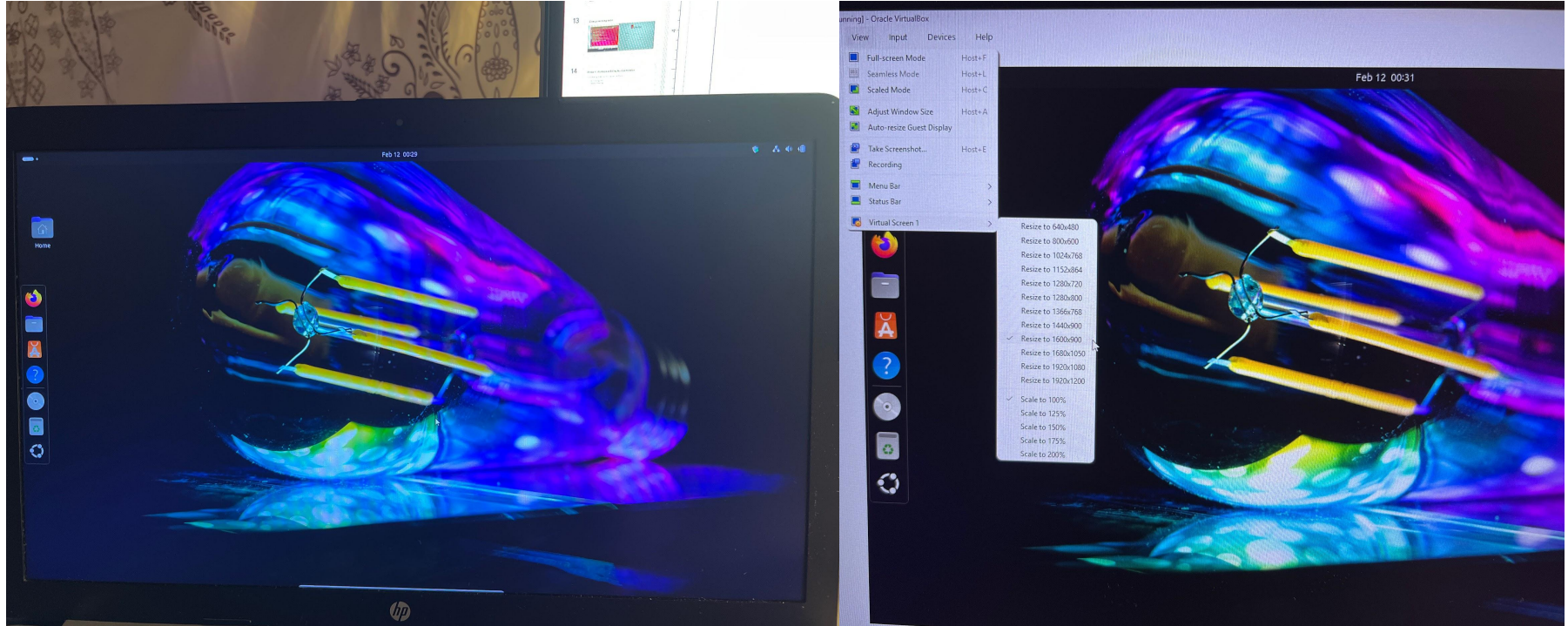
- Driver Integration
- Display Mapping

Driver Integration



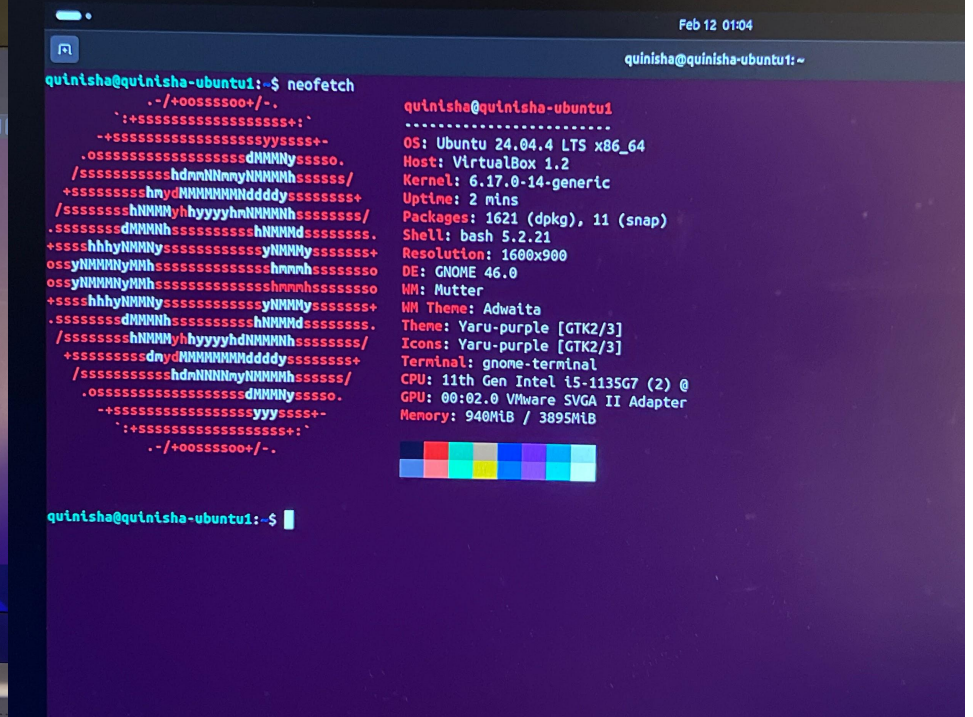
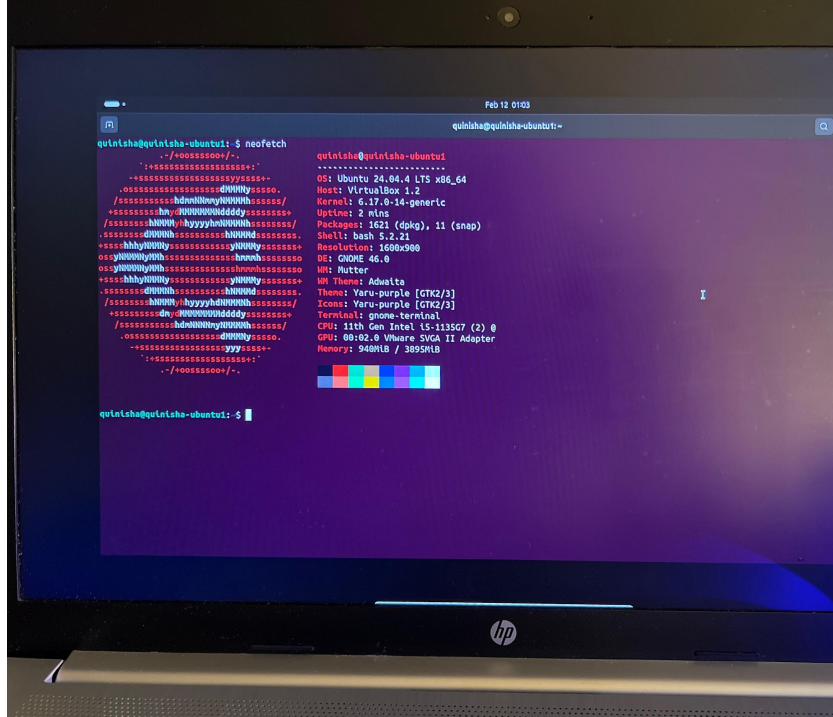
Installed VirtualBox Guest Additions to enable bidirectional clipboard support and advanced graphics handling

Display Mapping



Corrected initial scaling and "letterboxing" by manually configuring a 1600x900 resolution to optimize UI element visibility and provide a more comfortable viewing experience within the virtualized environment

Verification & Project Completion



Successfully executed a final system audit to verify the operational state of the environment